Exact nonequilibrium steady state of collective group-flip dynamics on quenched interaction neighborhoods

Pham Le Dat Nguyen[1, *]

[1] *Independent Researcher, Ha Noi, Viet Nam*

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Group-update models of opinion and spin dynamics are usually analyzed on annealed interaction structures, where the interacting group is redrawn at every step, and they relax to absorbing consensus. We study a two-state group-flip dynamics in which a firing node and every node that shares its state inside its fixed (quenched) graph-metric neighborhood switch state simultaneously. The dynamics has no absorbing state and no external noise, so its long-time behavior is a genuine nonequilibrium steady state. If every neighborhood is a strict majority of the system (R1), every complement of a neighborhood lies inside the neighborhood of each of its own nodes (R2), and all neighborhoods have β nodes, we prove that the stationary majority fraction is $\mathbb{E}[M] = (2-q)/(3-2q)$ with $q = \beta/n$, exactly for every system size n and independently of dimension and topology; equivalently, the mean absolute magnetization is $1/(3-2q)$. The proof rests on an exact affine closure for the mean minority size. Under R1 alone we prove that $\mathbb{E}[M]$ equals the fraction of updates that reverse the sign of the magnetization, and that the steady state violates detailed balance through one-way minority-erasure transitions, whose rate is $n(1-q)/(3-2q)$ under R1 and R2. For unequal neighborhoods the closure survives with one effective parameter and gives exact two-sided bounds. An annealed version with the same neighborhood size orders less in every case computed with $n - \beta \geq 2$ (up to $n = 3200$), so the exact steady state is a property of quenched structure. Below the R2 threshold, exact solutions of all 17 ring cases with $n \leq 33$ lie below the closed form, which we conjecture to be an upper bound there.

I. INTRODUCTION

Stochastic models of interacting two-state units, such as the voter model and majority-rule dynamics, are standard testing grounds of nonequilibrium statistical physics [1–6]. On a finite connected graph the voter model reaches one of its two absorbing consensus states, and a nontrivial stationary state requires an additional mechanism, typically spontaneous state changes (noise), as in the noisy voter or Kirman model [7–9]. Updates that involve groups of units, such as majority rule and nonlinear q-voter rules, introduce nonlinear drift terms that are absent from dyadic copying [4, 10–13], and group interactions are now studied systematically on higher-order structures [14, 15].

Many analytical results for group updates rely on annealed or mean-field structures, in which the interacting group is redrawn at every step, so that a mean-field drift can be written. A recent example is the group-driven voter model (GVM) of Kim *et al.* [16]: on annealed uniform hypergraphs the time to consensus grows as $A \ln N$, and for hyperedges larger than three it is shortest at an intermediate nonlinearity. Annealing is also a common route to analytical treatments of pairwise voter dynamics on complex networks [9, 17]. Two questions are then natural. Can a group-update dynamics on a *quenched* structure be solved exactly? And if it can, does the quenched structure change the macroscopic state, or does an annealed description with the same group size reproduce it?

Here we answer both questions for a simple rule that we call group-flip dynamics. Each node fires at rate one; when node i fires, the node and every node that shares its state inside a fixed metric ball $B_i(r)$ switch state together. Two features distinguish it from the models above. First, the update moves an entire same-state cluster of a neighborhood at once and involves no copying of a neighbor's state and no unanimity condition. Second, no state is absorbing: from consensus, the next fire creates a minority. The long-time state is therefore a stationary state without external noise, and we show that it violates detailed balance. The cluster move is reminiscent of cluster Monte Carlo algorithms [18], with the difference that there clusters are built to preserve a Boltzmann measure, whereas here the move is unconditional and the stationary measure is not reversible.

Our results are as follows. (i) Under a strict-majority condition R1 on the balls, the sign of the magnetization reverses exactly when a majority node fires; the stationary majority fraction $\mathbb{E}[M]$ therefore equals the fraction of updates that reverse the sign, and the dynamics contains one-way transitions, so the steady state is a nonequilibrium steady state (NESS) (Sec. III). (ii) If, in addition, every residual (the complement of a ball) lies inside the ball of each of its own nodes (R2) and all balls have the same size β, then

$$\mathbb{E}[M] = \frac{2-q}{3-2q}, \qquad q = \frac{\beta}{n}, \tag{1}$$

exactly for every finite n and for every geometry satisfying the conditions (Sec. IV). The derivation rests on an exact affine closure of the conditional mean minority size, so that no moment-closure approximation is involved.

* phmldtngyn.777@gmail.com

Related observables (mean absolute magnetization, rotation rate, erasure current, cycle-length law) follow in closed form, and for unequal balls the closure survives with a single effective parameter and yields exact two-sided bounds. (iii) An annealed version with the same ball size, solved exactly as a Markov chain on the magnetization, orders less than the quenched system in every case computed with $n - \beta \geq 2$, up to $n = 3200$; the gap does not close as n grows in that range. Equation (1) is thus a property of quenched neighborhoods, and the exact lemmas identify the mechanism (Sec. V). (iv) Below the R2 threshold, minorities are no longer erased in one update. We give the one-step extinction probability in closed form on rings, solve the stationary chain for all 17 ring cases with $n \leq 33$, and find that all values lie below Eq. (1), which we state as a conjecture (Sec. VI). Table I places the model among related dynamics.

II. MODEL

A. Quenched neighborhoods

Let $G = (V, E)$ be a connected graph with n nodes and graph distance d. For a radius r, the interaction neighborhood of node i is the ball $B_i = \{j : d(i, j) \leq r\}$, of size $\beta_i = |B_i|$; when all balls have the same size we write β and define the coverage $q = \beta/n$. The balls are quenched: they are fixed for the whole dynamics. Since d is symmetric, $j \in B_i \iff i \in B_j$, a property used repeatedly below. The complement $R_i = V \setminus B_i$ is the residual of i.

Our main example is the ring $C_n(1, 2)$, in which each node is linked to its first and second neighbors on a cycle. Its distance is $d(i, j) = \lceil \delta(i, j)/2 \rceil$, with δ the circular lattice distance, so the balls are arcs of $\beta = \min(n, 4r+1)$ nodes. In two dimensions we use the analogous square lattice with first- and second-neighbor links along each axis, for which $\beta = 8r^2 + 1$ without wrap-around [19].

Remark 1 (Origin of the construction). These lattices are the 2-section graphs of uniform hypergrids whose hyperedges are three consecutive nodes along an axis, the formulation used in an earlier version of this work [20]. The dynamics depends on a hypergraph only through its 2-section graph, so no higher-order interaction beyond the pairwise projection is involved, and we use graph language throughout.

B. Group-flip dynamics

Each node carries a state $\sigma_i \in \{0, 1\}$ and fires at rate one, independently. When node i fires, every node $j \in B_i$ with $\sigma_j = \sigma_i$ switches to $1 - \sigma_i$ simultaneously (Fig. 1). A single update therefore moves the same-state part of a ball, and its outcome is deterministic given the firing node. The total update rate is n in every configuration. The version with $Q > 2$ states, in which the moved class adopts a state chosen uniformly among the other $Q - 1$, is solved exactly in the well-mixed limit $B_i = V$ in the Supplemental Material [19]; all results below are for $Q = 2$.

With n_σ the number of nodes in state σ, the magnetization is $m = (n_1 - n_0)/n$ and the majority fraction is $M = \max(n_0, n_1)/n = (1 + |m|)/2$. The primary observable is the stationary (long-run time-averaged) value $\mathbb{E}[M]$. A rotation is an update that changes the majority label, i.e., reverses the sign of m. We denote by K the number of updates between consecutive rotations and by T the corresponding time; a cycle is the interval between consecutive rotations.

C. Geometric conditions

Two conditions organize the analysis:

R1:: $\beta_i > n/2$ for all i: every ball is a strict majority of the system;

R2:: $\mathrm{diam}(R_i) \leq r$ for all i: every residual lies inside the ball of each of its own nodes.

R2 compares two lengths: the diameter of the largest minority that a single update from consensus can create (a residual) and the interaction range r. On the ring [19],

$$\text{R1} \iff 2\beta > n, \qquad \text{R2} \iff \beta = n \text{ or } 3\beta \geq 2n - 1, \tag{2}$$

so that R1∧R2 holds asymptotically for $q \geq 2/3$ (Region I), R1 alone for $1/2 < q < 2/3$ (Region II), and neither for $q \leq 1/2$ (Region III); we use the exact finite-n forms throughout.

Because R1 requires $\beta_i > n/2$, the radius grows with n at fixed q, as $r = \Theta(n^{1/d})$ on d-dimensional lattices [19]. Each update thus acts on a finite fraction of the system, as in mean-field treatments, but on a set that is fixed in space. The comparison with an annealed version at the same β in Sec. V isolates the effect of this structure from the effect of the large neighborhood size.

III. EXACT RESULTS UNDER R1: SIGN REVERSALS AND BROKEN DETAILED BALANCE

Theorem 1 (Rotation identity)**.** *Assume R1 (equal balls are not required). From any state with a strict majority $\mathcal{M}$ and minority $\mathcal{A}$, an update is a rotation if and only if the firing node belongs to $\mathcal{M}$; the instantaneous rotation rate is $|\mathcal{M}| = nM$. Tie states are left at the first update and never re-entered, and carry zero stationary mass.*

Proof. Let the firing node be $i \in \mathcal{M}$. After the update the class of the new state is $\mathcal{A} \cup (\mathcal{M} \cap B_i)$ and that of the old state is $\mathcal{M} \setminus B_i$. Their size difference is

$$|\mathcal{A}| + |\mathcal{M} \cap B_i| - |\mathcal{M} \setminus B_i| = (2\beta_i - n) + 2|\mathcal{A} \setminus B_i| > 0, \tag{3}$$

TABLE I. Group-flip dynamics compared with related two-state dynamics. The entries describe the settings analyzed in the cited works; "long-time state" refers to a finite system.

Model	Interaction structure analyzed	Update	Long-time state
Voter model [1, 2]	graph complete graph;	node copies a random neighbor copying plus	absorbing consensus stationary,
Noisy voter model [7–9]	networks (annealed approx.)	spontaneous flips	sustained by noise
Majority rule [4, 13]	mean field; hypergraphs	group adopts its local majority node adopts a	absorbing consensus
Group-driven voter [16]	annealed uniform hypergraphs	unanimous sample same-state part of a	absorbing consensus stationary without noise;
This work	quenched metric balls	ball switches	exact $\mathbb{E}[M]$

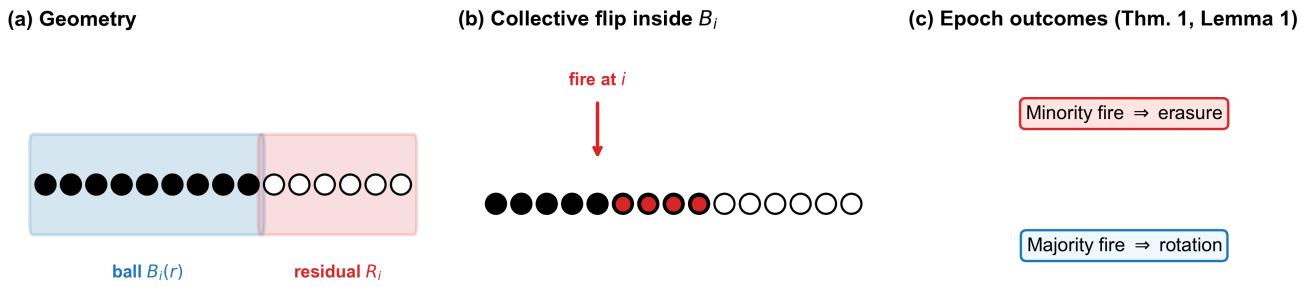


FIG. 1. Group-flip update on a ring. (a) Ball $B_i(r)$ (shaded) and residual R_i. (b) A fire at i: all nodes of i's state inside B_i (red) switch together. (c) The two outcomes from a strict-majority state: a majority fire reverses the majority label (Theorem 1); a minority fire from a self-contained minority erases it (Lemma 1).

so the majority label changes. If $i \in \mathcal{A}$, the minority can only shrink, and no rotation occurs. Each majority node fires at rate one. From a tie, Eq. (3) with $|\mathcal{M}| = |\mathcal{A}|$ shows that every update produces a strict majority, and strict-majority states are mapped to strict-majority states. □

Corollary 1 (Static–dynamic identity)**.** *Under R1, the long-run fraction of updates that reverse the sign of m equals $\mathbb{E}[M]$, the long-run rotation rate per unit time is $n\mathbb{E}[M]$, and $\mathbb{E}[K] = 1/\mathbb{E}[M]$.*

Proof. Firing epochs form a Poisson process of rate n, and the firing node is uniform and independent of the configuration. By the PASTA property applied to the firing epochs, the configuration seen by a fire has the stationary law, so a fire is a rotation with probability $\mathbb{E}[M]$ by Theorem 1. If the chain has several closed classes, the statement holds class by class [19]. □

Corollary 1 gives $\mathbb{E}[M]$ a second, dynamical meaning: it is the probability that a randomly chosen update reverses the sign of the order parameter. Since $\mathbb{E}[M] > 1/2$, the sign of m reverses at a rate of order n per unit time at every system size, while $|m|$ stays of order one. The order is therefore an order in $|m|$ without a persistent sign, in contrast with ordered phases in which reversals of the magnetization become rarer as the system grows.

Proposition 1 (Broken detailed balance)**.** *Assume R1. Let x be a configuration whose minority $\mathcal{A}$ is nonempty and let a fire of a node $j \in \mathcal{A}$ with $\mathcal{A} \subseteq B_j$ map x to a consensus configuration y. Then no single update maps y to x. Consequently, whenever such erasure transitions occur with positive stationary rate, the stationary dynamics is not reversible: the steady state carries nonzero probability currents [21].*

Proof. From a consensus configuration, a fire at k switches exactly B_k, so the configurations reachable in one step are those in which the nonconsensus state occupies a ball. In x that state occupies $\mathcal{A}$, and $|\mathcal{A}| < n/2 < \beta_k$ for every k by R1. A reversible Markov chain requires $w(x \to y) > 0 \iff w(y \to x) > 0$ for all pairs; this fails. □

By the same PASTA argument, the stationary rate of minority fires is $n(1 - \mathbb{E}[M])$. Where every minority fire is an erasure, as in Region I (Lemma 1 below), this is the rate of one-way transitions; Eq. (9) gives it in closed form.

IV. EXACT STEADY STATE UNDER R1 AND R2

A. Epoch chain and self-containment

Let A_m be the minority set immediately after the mth rotation, $a_m = |A_m|$, and i_m the node whose fire produced that rotation.

Lemma 1 (Epoch invariant)**.** *Under R1 and R2, $A_m \subseteq R_{i_m}$ for every m. Consequently $A_m \subseteq B_j$ for every $j \in A_m$ (self-containment), so any minority fire erases the whole minority, and $a_m \le n - \beta_{i_m}$. Equal balls are not required.*

Proof. From consensus a fire at i switches B_i, which is the new majority by R1, so the new minority is R_i. Suppose $A \subseteq R_i$ is the current minority. R2 gives $R_i \subseteq B_j$ for every $j \in R_i$, hence $A \subseteq B_j$ for every $j \in A$. If the next fire is at $j \in A$, the whole minority switches, the system reaches consensus, and the following fire, at some i', creates the minority $R_{i'}$ and a rotation. If the next fire is at $i \in \mathcal{M}$, only majority nodes inside B_i switch (minority nodes in B_i already carry the target state); a rotation occurs (Theorem 1) and the new minority is $\mathcal{M} \setminus B_i \subseteq R_i$. Both branches preserve the invariant. □

Lemma 1 is where the quenched geometry enters: every minority is a compact domain contained in a residual, and it is erased in one update by any of its own nodes.

B. Exact affine closure

Lemma 2 (Overlap identity)**.** *Under R1, R2 and equal balls, for i uniform in the majority $\mathcal{M}$ and a minority A of size a,*

$$\mathbb{E}\big[|A \cap B_i| \,\big|\, a\big] = \frac{a(\beta - a)}{n - a}. \tag{4}$$

Proof. Exchange the order of summation and use $j \in B_i \iff i \in B_j$: $\sum_{i\in\mathcal{M}} |A \cap B_i| = \sum_{j\in A} |B_j \cap \mathcal{M}|$. By self-containment $B_j \supseteq A$, so $|B_j \cap \mathcal{M}| = \beta - a$. Divide $a(\beta - a)$ by $|\mathcal{M}| = n - a$. □

A minority fire has probability a/n; it erases the minority, and the following fire creates a minority of size $n - \beta$. A majority fire has probability $1 - a/n$ and creates the minority $\mathcal{M} \setminus B_i$, of size $n - \beta - a + |A \cap B_i|$. By Lemma 2,

$$\mathbb{E}[a' \mid a,\ \text{majority fire}] = \frac{(n - \beta)(n - 2a)}{n - a}, \tag{5}$$

and weighting the two branches gives

$$\mathbb{E}[a_{m+1} \mid a_m = a] = (1 - q)(n - a). \tag{6}$$

The conditional mean is affine in a_m, so taking expectations against any distribution of a_m produces no correction from higher moments. This is the step on which the exact result rests.

C. Stationary state

The epoch sets form a Markov chain on a finite family of subsets. Let π be any invariant distribution and $\bar{x} = \mathbb{E}_\pi[a]/n$; uniqueness is not needed. Taking π-expectations in Eq. (6) gives $\bar{x} = (1 - q)(1 - \bar{x})$, hence

$$\bar{x} = \frac{1 - q}{2 - q} \tag{7}$$

for every invariant distribution. A cycle contains a second update if and only if its first update is a minority fire, so $K \in \{1, 2\}$ and $P(K = 2) = \bar{x}$.

Theorem 2 (Exact stationary majority fraction)**.** *Assume $Q = 2$, R1, R2, and equal balls β. Then, for every finite n and every initial condition,*

$$\mathbb{E}[M] = \frac{2 - q}{3 - 2q}, \qquad q = \frac{\beta}{n}. \tag{8}$$

Proof. Inter-update times are exponential with mean $1/n$ and independent of the sequence of firing nodes. During the first interval of a cycle the majority fraction is $1 - a_m/n$; if a second interval occurs (probability a_m/n), the system is in consensus and $M = 1$. The expected time integral of M over a cycle is therefore $n^{-1}(1-\bar{x})+n^{-1}\bar{x} = 1/n$, and the expected cycle length is $(1 + \bar{x})/n$. Cycles are stationary but not independent; the ergodic theorem applied to the stationary epoch chain gives the ratio of these means on each closed class, and Eq. (7) makes it the same on every class (Appendix A). Hence $\mathbb{E}[M] = 1/(1 + \bar{x}) = (2 - q)/(3 - 2q)$. □

Theorem 2 applies on the R1∧R2 domain, where $\mathbb{E}[M]$ ranges from $(4n + 1)/(5n + 2)$ at the ring boundary $3\beta = 2n - 1$ (tending to 4/5) to 1 at $q = 1$. Values of the closed form at smaller q are used below only as a comparator. The result does not depend on the dimension or topology of G: two geometries with the same q that satisfy R1 and R2 have the same stationary majority fraction. Native-ball simulations of rings ($n = 400, 401$) and two-dimensional lattices ($L = 12, 30$), for which R1 and R2 were verified by breadth-first search, reproduce Eq. (8) within 2.1×10^{-5} (largest $|z| = 0.8$) [19].

Corollary 2 (Closed-form steady-state observables)**.** *Under the assumptions of Theorem 2,*

$$\begin{aligned} \mathbb{E}|m| &= \frac{1}{3 - 2q}, & \mathbb{E}[K] &= \frac{3 - 2q}{2 - q}, \\ J_{\text{rot}} &= n\,\frac{2 - q}{3 - 2q}, & P(K = 2) &= \frac{1 - q}{2 - q}, \\ J_{\text{erase}} &= n\,\frac{1 - q}{3 - 2q}, & \mathrm{CV}_T^2 &= \frac{7 - 8q + 2q^2}{(3 - 2q)^2}, \end{aligned} \tag{9}$$

where J_{rot} is the long-run rotation rate per unit time, J_{erase} the long-run rate of one-way erasure transitions (Proposition 1), and CV_T^2 the squared coefficient of variation of the cycle time.

Proof. $|m| = 2M - 1$. $\mathbb{E}[K]$ and $P(K = 2)$ follow from Eq. (7); $J_{\text{rot}} = n\mathbb{E}[M]$ by Corollary 1; every minority fire is an erasure by Lemma 1, so $J_{\text{erase}} = n(1 - \mathbb{E}[M])$. For T, write $T = G_1 + BG_2$ with $G_{1,2}$ exponential of mean $1/n$ and B Bernoulli of mean $\bar{x}$, independent of G_1, G_2 [19]. □

The cycle statistics $\mathbb{E}[K]$ and CV_T^2 test the cycle decomposition independently of $\mathbb{E}[M]$; in simulations of the same Region-I geometries they agree with Eq. (9) with largest $|z| = 1.9$ [19]. The erasure current J_{erase} is strictly positive for $q < 1$, so by Proposition 1 the steady state of Theorem 2 is a NESS for every $q < 1$.

D. Unequal balls

Rotation and self-containment do not use equal balls; only Lemma 2 does. For unequal balls the same exchange of summation gives $\sum_{i\in\mathcal{M}} |A \cap B_i| = S_A - a^2$ with $S_A = \sum_{j\in A} \beta_j$, and the conditional mean becomes

$$\mathbb{E}[a' \mid A] = n - \bar{\beta} - a\Big(1 + \frac{\bar{\beta}}{n}\Big) + \frac{2S_A}{n}, \tag{10}$$

with $\bar{\beta} = n^{-1} \sum_i \beta_i$. This is affine in a and in the single additional statistic S_A, so the closure survives with one effective parameter.

Proposition 2 (Unequal balls)**.** *Assume $Q = 2$, R1 and R2, $\beta_i < n$ for some i, and let $\bar{q} = \bar{\beta}/n$, $q_{\min} = \min_i \beta_i/n$, $q_{\max} = \max_i \beta_i/n$. On each closed class c of the epoch chain, with stationary law π_c, define $q_c^* = \mathbb{E}_{\pi_c}[S_A]/(n\mathbb{E}_{\pi_c}[a]) \in [q_{\min}, q_{\max}]$. Then*

$$\mathbb{E}_c[M] = g(q_c^*), \qquad g(s) = \frac{2 + \bar{q} - 2s}{3 - 2s}, \tag{11}$$

and, since g is decreasing,

$$g(q_{\max}) \le \mathbb{E}[M] \le g(q_{\min}). \tag{12}$$

For equal balls, $q^* = \bar{q} = q$ and Eq. (11) reduces to Eq. (8).

Proof. Averaging Eq. (10) over π_c gives $\bar{x}_c = (1-\bar{q})/(2+\bar{q}-2q_c^*)$. The cycle structure ($K \in \{1, 2\}$, $P(K = 2 \mid a) = a/n$) and the renewal–reward argument of Theorem 2 are unchanged, so $\mathbb{E}_c[M] = 1/(1 + \bar{x}_c)$. Finally $g'(s) = -2(1 - \bar{q})/(3 - 2s)^2 \le 0$. Full proof in [19]. □

We checked Eq. (11) on 60 distinct ball systems with unequal ball sizes satisfying R1 and R2 (rings with $9 \le n \le 14$ and up to five random chords, $r \in \{2, 3, 4\}$), computing q^* directly from the rotation-weighted epoch law of the exact stationary chain rather than inferring it from $\mathbb{E}[M]$: the identity holds to 1.2×10^{-15}, and the bounds (12) hold in every case [19]. In all 60 cases $q^* < \bar{q}$, i.e., minorities sit preferentially on nodes with smaller balls, and therefore $\mathbb{E}[M] > (2 - \bar{q})/(3 - 2\bar{q})$, by between 0.0008 and 0.013. We do not have a proof that $q^* \le \bar{q}$ in general.

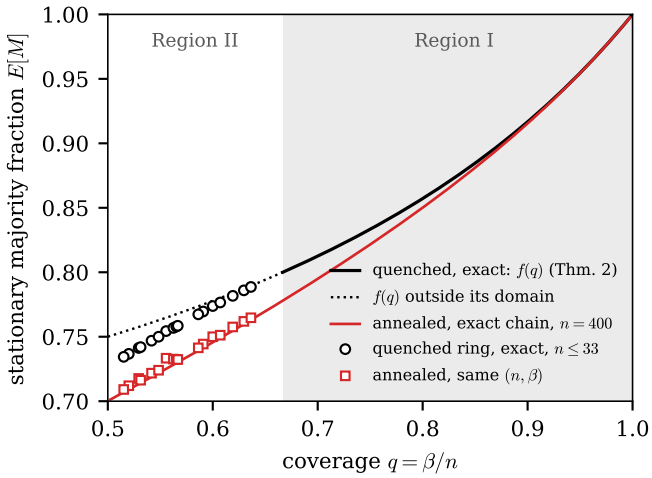


FIG. 2. Quenched versus annealed neighborhoods of equal size. Solid black: exact result Eq. (8), valid for any geometry satisfying R1 and R2 (shaded, asymptotic location $q \ge 2/3$); dotted: the same function outside its domain, used as a comparator. Circles: exact stationary values on the quenched ring in Region II (Table II). Squares: annealed version with the same (n, β). Red line: annealed version at $n = 400$, every integer β, exact chain.

V. QUENCHED VERSUS ANNEALED NEIGHBORHOODS

To separate the effect of quenched structure from that of neighborhood size, consider an annealed version: at every fire of i the interaction set is $\{i\}$ together with $\beta - 1$ nodes drawn uniformly without replacement from $V \setminus \{i\}$. The number of switched nodes is then $1 + h$ with h hypergeometric, and the dynamics reduces to a Markov chain on $n_1 \in \{0, \dots, n\}$, whose stationary law we compute by a direct linear solve [19].

Figure 2 compares the two. In Region II the annealed value lies below the exact quenched ring value in all 17 cases of Table II, by between 0.021 and 0.026. In Region I, at $n = 400$, the annealed value lies below Eq. (8) for every $\beta \le n - 2$, by up to 0.022; at $\beta = n - 1$ the two coincide, since each interaction set then misses a single node. The gap does not close with system size in the range computed: at $q = 0.7$ it is 0.0173, 0.0175, 0.0176, 0.0177, 0.0177 and 0.0178 for $n = 100, 200, 400, 800, 1600$ and 3200, and at $q = 0.8$ it grows from 0.0068 ($n = 100$) to 0.0071 ($n = 1600$) [19]. These are exact finite-n values; we do not prove a large-n limit.

The exact lemmas identify where the difference comes from. Write $x = a/n$ for the minority fraction. In the quenched system under R1 and R2, a minority fire erases the minority ($x \to 0$), and a majority fire leaves, on average, $x' = (1 - q)(1 - 2x)/(1 - x)$ by Eq. (5). In the annealed system at large n, hypergeometric concentration gives $x \to (1 - q)x$ for a minority fire and $x \to (1-q)(1-x)$ for a majority fire. Both quenched branches leave a smaller minority, since $(1-2x)/(1-x) < 1-x$ for

$0 < x < 1/2$. The origin is self-containment: a minority is a compact domain whose nodes lie in one another's balls, so it is erased completely by a single fire from inside, and it is under-represented in the balls of majority nodes [Eq. (4)], which therefore leave fewer minority nodes behind. An annealed interaction set carries no such correlation with the minority. Iterating the annealed large-n map reproduces the exact annealed chain at the largest sizes computed (at $q = 0.7$: 0.79472 ± 0.00002 from the map, 0.79475 for the chain at $n = 3200$) [19]; this map is a heuristic for the annealed system, whereas the quenched statements are exact.

In pairwise voter dynamics, annealing the long-range links of a small-world network preserves much of the phenomenology [17]. Here, by contrast, annealing at fixed neighborhood size changes the stationary order parameter at order one in n within the computed range: the exact steady state is a property of the quenched geometry, not of the neighborhood size alone.

VI. BEYOND ONE-STEP ERASURE

A. One-step extinction probability

When R2 fails, a newly created minority is not necessarily erased by the first fire of one of its nodes. For residuals R_i define

$$c_G = \frac{1}{n^2} \sum_{i,j} \mathbf{1}\{R_i \subseteq B_j\}, \tag{13}$$

$$\tilde{c} = \frac{\sum_i \sum_{j \in R_i} \mathbf{1}\{R_i \subseteq B_j\}}{\sum_i |R_i|}, \tag{14}$$

with $\tilde{c} = 1$ if all residuals are empty. c_G is a purely geometric overlap statistic. $\tilde{c}$ is the probability that a newly created minority (a residual) is erased by the first fire of one of its own nodes, because that fire is uniform in the residual [19]; under R2, $\tilde{c} = 1$.

Proposition 3 (Ring containment)**.** *On the ring with* $\beta = \min(n, 4r+1)$, $c_G = \min\{1, \max(0, 2\beta - n + 1)/n\}$ *and*

$$\tilde{c} = \begin{cases} 0, & 2\beta < n, \\ \dfrac{2\beta - n + 1}{n - \beta}, & 2\beta \geq n,\ 3\beta < 2n - 1, \\ 1, & 3\beta \geq 2n - 1, \end{cases} \tag{15}$$

continuous at $3\beta = 2n - 1$.

Proof. For $\beta < n$, a ball arc of length β contains the residual arc of length $n - \beta$ if and only if its center lies in an interval of $2\beta - n + 1$ positions; for $\tilde{c}$ intersect this interval with the residual. At $2\beta = n$ the residual is itself a ball and exactly one node of it is admissible. Details in [19]. □

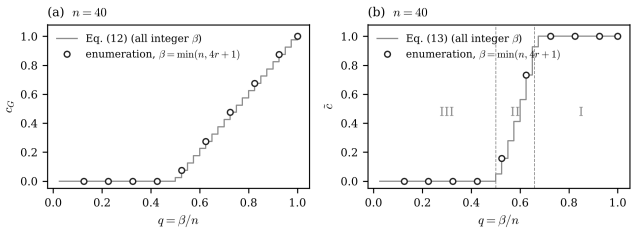


FIG. 3. Ring containment for $n = 40$. (a) c_G and (b) $\tilde{c}$ against q: closed forms (Proposition 3) at every integer β (steps) and exhaustive enumeration at the realized $\beta = \min(n, 4r+1)$ (circles). Dashed lines in (b): $\beta = n/2$ and $\beta = (2n-1)/3$.

Equation (15) agrees exactly, in rational arithmetic, with exhaustive enumeration in all 192 cases (n, r) with $12 \leq n \leq 40$ (Fig. 3). At the Region I/II boundary one-step extinction is lost: $\tilde{c}$ falls continuously below 1 once the residual diameter exceeds the interaction range, and it vanishes when R1 fails.

B. Exact finite-size steady states in Region II

For the ring in Region II we solved the stationary two-state chain on all configurations reachable from consensus. For the six odd $n \leq 25$ the reachable minority sets form a single irreducible class (73 to 51 201 sets), the chain lumps exactly under the dihedral group and state exchange, and the stationary values were obtained in exact rational arithmetic and confirmed by an independent floating-point solver of the unreduced chain (residual $\leq 4.1 \times 10^{-14}$). A third, independent solver extends the computation to all 17 Region-II cases with $n \leq 33$, odd and even (residual $< 5 \times 10^{-13}$; Table II).

All 17 values lie below the closed form, by between 0.0009 and 0.0195, and above the annealed values.

Conjecture 1. *In Region II of the ring,* $\mathbb{E}[M] \leq (2 - q)/(3 - 2q)$ *for all* n *and all admissible* r.

By Corollary 1, which needs only R1, the conjecture is equivalent to $\mathbb{E}[K] \geq (3 - 2q)/(2 - q)$: below the R2 threshold, sign reversals of m would be rarer per update than in the exactly solvable regime. Across the 17 cases the deviation increases monotonically with q, from -0.0195 at $q = 0.515$ to -0.0009 at $q = 0.636$, whereas $\tilde{c}$ does not (it is $1/2$ at $q = 0.556$ and $2/5$ at $q = 0.565$); within this set, q organizes the deviation better than the one-step quantity $\tilde{c}$.

Interpretation. In Region I the largest minority that a single update can create has diameter at most the interaction range, and the closure of Sec. IV is exact. In Region II a minority can survive the first fire from inside it, possibly as several fragments, and longer-lived minorities lower the time-averaged majority fraction. The negative deviations are consistent with this picture, but the exact chain gives the stationary mean, not a decomposition by

TABLE II. All Region-II cases of the ring with $n \le 33$ ($n/2 < \beta < (2n-1)/3$, $\beta = 4r+1$). States: configurations reachable from consensus, up to state exchange. $\mathbb{E}[M]$: exact stationary value of the quenched dynamics; for the six odd $n \le 25$ the exact rational is given. $f(q) = (2-q)/(3-2q)$ is evaluated outside its domain as a comparator. $\mathbb{E}[M]_{\text{ann}}$: annealed version with the same β.

n	r	β	q	$\tilde{c}$	states	$\mathbb{E}[M]$	$f(q)$	$\mathbb{E}[M]-f(q)$	$\mathbb{E}[M]_{\text{ann}}$
9	1	5	0.5556	1/2	73	0.754386 (43/57)	0.764706	−0.0103	0.733333
15	2	9	0.6000	2/3	481	0.773810 (65/84)	0.777778	−0.0040	0.750000
16	2	9	0.5625	3/7	1025	0.756831	0.766667	−0.0098	0.732639
17	2	9	0.5294	1/4	2177	0.741294 (149/201)	0.757576	−0.0163	0.717647
21	3	13	0.6190	3/4	2689	0.781711 (265/339)	0.783784	−0.0021	0.757576
22	3	13	0.5909	5/9	5633	0.769553	0.775000	−0.0054	0.744318
23	3	13	0.5652	2/5	11777	0.757895 (72/95)	0.767442	−0.0095	0.732441
24	3	13	0.5417	3/11	24577	0.746914	0.760870	−0.0140	0.721726
25	3	13	0.5200	1/6	51201	0.736721 (319/433)	0.755102	−0.0184	0.712000
27	4	17	0.6296	4/5	13825	0.785965	0.787234	−0.0013	0.761905
28	4	17	0.6071	7/11	28673	0.776560	0.780000	−0.0034	0.751190
29	4	17	0.5862	1/2	59393	0.767367	0.773585	−0.0062	0.741379
30	4	17	0.5667	5/13	122881	0.758488	0.767857	−0.0094	0.732353
31	4	17	0.5484	2/7	253953	0.750000	0.762712	−0.0127	0.724014
32	4	17	0.5312	1/5	524289	0.741958	0.758065	−0.0161	0.716283
33	4	17	0.5152	1/8	1081345	0.734396	0.753846	−0.0195	0.709091
33	5	21	0.6364	5/6	67585	0.788618	0.789474	−0.0009	0.764706

mechanism, and we do not isolate multi-step survival as the cause.

C. Exploratory test across graph families

At intermediate coverage, simulations of four graph families (rings, two-dimensional lattices with eight neighbors, configuration-model and Watts–Strogatz graphs) show family-dependent stationary values at equal (n, q). We tested whether $\tilde{c}$ accounts for this dependence with a weighted regression on 216 runs, using interaction sets truncated to a common β. It does not: the coefficient of $\tilde{c}$ is negative, -0.0196 [cluster-bootstrap 95% interval $(-0.0222, -0.0169)$], and including $\tilde{c}$ increases the separation between families. Because $\tilde{c}$ varies almost only within the ring family, the test does not identify an alternative mechanism. The design, the analysis record and all sensitivity analyses are given in the Supplemental Material [19].

VII. DISCUSSION

Group-flip dynamics on quenched neighborhoods provides an exactly solvable nonequilibrium steady state without external noise. Under R1 and R2 with equal balls the stationary majority fraction, the mean absolute magnetization, the rotation and erasure currents and the cycle law are known in closed form for every finite n, and they depend on the geometry only through the coverage q. The solution does not use a moment closure: the conditional mean minority size is exactly affine, because self-containment turns the overlap between a minority and a random majority ball into a function of the minority size alone [Eq. (4)]. The same mechanism explains why an annealed description with the same group size fails: it discards the correlation between minority domains and neighborhoods that the quenched geometry creates.

Relation to annealed group dynamics. Kim *et al.* [16] obtain leading-order prefactors of a logarithmic exit time for a unanimity-based rule on annealed hypergraphs, with an optimal nonlinearity that results from a competition between diffusive and drift-dominated behavior. Their edge-update variant also changes several nodes at once, but it copies a unanimous state and keeps consensus absorbing. Here the moved set is a same-state part of a quenched ball, consensus is not absorbing, and the object computed is a stationary value rather than an exit time. The two results therefore address different questions; the annealed comparison of Sec. V shows that, for the group-flip rule, replacing the quenched balls by annealed sets of the same size changes the stationary state.

Limitations. (i) All exact results are for two states. (ii) The closed form (8) requires equal balls; for unequal balls Proposition 2 gives an exact expression with one effective parameter and exact bounds. (iii) R1 forces the radius to grow as $n^{1/d}$, so the large-n limit at fixed q is not a short-range limit; Theorem 2 is a finite-n statement about extensive, spatially fixed neighborhoods. (iv) Region II is solved only numerically, for $n \le 33$, and Conjecture 1 is unproven. (v) The annealed size dependence

is computed, not proved.

Open problems. A theory of Region II must describe minorities that survive their first erasure attempt, for instance through multi-step containment probabilities or the statistics of minority fragments. The fluctuations of M are not fixed by the present closure, since they involve pairwise overlaps of balls. A proof that $q^* \le \bar{q}$ for unequal balls would turn the observation of Sec. IV into a bound. Finally, whether a short-range version of group-flip dynamics retains an ordered, sign-reversing steady state is open.

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DATA AVAILABILITY

The code and data that support the findings of this article are openly available [20], at https://doi.org/10.5281/zenodo.24050030 and https://github.com/phmldtngyn/Group-flip-consensus-on-hypergraphs.

Appendix A: Ergodic step in Theorem 2

Start the epoch chain in an invariant distribution π, which exists because the state space is finite. Let $Y_m = \int_{\text{cycle } m} M\,dt$ and τ_m be the reward and length of cycle m. The sequence (Y_m, τ_m) is stationary as a functional of a stationary chain; it is not independent, and independence is not used. By the Birkhoff ergodic theorem, $N^{-1}\sum_{m\le N} Y_m \to \mathbb{E}[Y \mid \mathcal{I}]$ and $N^{-1}\sum_{m\le N} \tau_m \to \mathbb{E}[\tau \mid \mathcal{I}] \ge 1/n$ almost surely, with $\mathcal{I}$ the invariant σ-field. On the event that the chain lies in a closed class c, these are the class means $\mathbb{E}_c[Y] = 1/n$ and $\mathbb{E}_c[\tau] = (1+\bar{x}_c)/n$. Applying Eq. (7) to the chain restricted to c gives $\bar{x}_c = (1-q)/(2-q)$ for every class, so the ratio is $1/(1+\bar{x})$ almost surely. From an arbitrary initial condition the epoch chain enters a closed class in finite time almost surely, so the long-run time average of M is the same for every initial law [19].

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